

CS110X Lab: Input-Output formats and Conditional statements

A: Input and output formatting.

1. Write a C program to read the following details of a student:

Name, Roll Number, Age, CGPA, Grade

Display the information in a neatly formatted manner.

Sample Output

```
Student Information
```

```
-----  
Name           : Alice  
Roll Number    : 101  
Age            : 19  
CGPA           : 8.75  
Grade          : A  
-----
```

2. Read a floating-point number and display it:
With 2 decimal places
With 4 decimal places
With 6 decimal places
3. Read an integer and display it using different field widths:
Width 5
Width 8
Left aligned (width 8)

Leading zeros (width 5)
Sample Output

```
Width 5      : |   25|
Width 8      : |      25|
Left Align   : |25      |
Leading Zero  : |00025|
```

4. **Question using format specifiers for scanf**

A student's record is entered in the following format:

101,Alice,8.75

Write a C program to read the Roll Number, Name, and CGPA using a single scanf() statement.

5. Write a C program to read a person's full name (including spaces between firstname lastname etc) and display it. The name must be stored inside a single variable.

6. Read the following information from one line:

101 A 8.75

where

Roll Number (integer)

Grade (character)

CGPA (float)

Display all values.

7. Travel Ticket Generator: The passenger information is entered in the following format:

Passenger Name,Train Number,Date,Seat Number,Fare

Example

Amit Kumar,12505,21-08-2026,S3-45,875.50

Generate the ticket

```
=====
                        RAILWAY TICKET
=====

Passenger : Amit Kumar
Train No. : 12505
Journey   : 21-08-2026
Seat No.  : S3-45
Fare      : Rs. 875.50

=====
```

8. Average of Three Numbers Problem Statement

Write a C program to read three integers and calculate their average.

Display the average without type casting. Display the average after using explicit type casting.

9. Read total marks of 5 students. Display percentage. Decide letter grade using ENUM. Print letter grade as per percentage. Letter grades are {A,B,C,D,F}. Decide your enum accordingly.

B: Introduction to conditionals.

Write a C program that takes a student's marks (0-100) as input and prints the corresponding grade according to the following rules:

- 90-100 → Grade A
- 75-89 → Grade B
- 60-74 → Grade C
- 40-59 → Grade D
- Below 40 → Fail

Sample Input/Output:

Input: 92 Output: Grade: A

```
Input: 63
Output: Grade: C
```

```
Input: 35
Output: Grade: Fail
```

Steps to solve the problem:

1. Declare an integer variable **marks** to store the students score.
2. Prompt the user to enter marks (between 0 and 100).
3. Read the input using **scanf**.
4. Use a series of **if--else if--else** conditions to check the range of marks:
 - If marks ≥ 90 and $\leq 100 \rightarrow$ print "Grade: A"
 - Else if marks $\geq 75 \rightarrow$ print "Grade: B"
 - Else if marks $\geq 60 \rightarrow$ print "Grade: C"
 - Else if marks $\geq 40 \rightarrow$ print "Grade: D"
 - Else $\rightarrow$ print "Grade: Fail"
5. End the program.

Follow-up Exercise: Grading Multiple Students

Extend the grading program so that it can take the marks of **5 students** and print their grades.

Sample Input:

```
Enter the marks of Student 1- 45
Enter the marks of Student 2- 75
.
.
.
.
.
```

Sample Output:

```
Grade of Student 1- D
Grade of Student 2- B
.
.
.
.
.
```

Steps to solve the problem:

1. Declare 5 separate integer variables, e.g., `marks1`, `marks2`, ..., `marks5`.
2. For each student:
 - Prompt the user to enter marks.
 - Read the input using `scanf`.
 - Use the same conditional logic (if–else ladder) to determine the grade.
 - Print the grade.
3. Repeat the above steps manually for all 5 students.
4. End the program.